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EXAMPLE 1.8

The voltage across a wire is (100 ± 5)V and the current passing through it is (10±0.2) A. Find the resistance of the wire.

Solution

Voltage V = (100±5)V

Current I = (10±0.2) A

Resistance R = ?

Then resistance R is given by Ohm's law,

$R = \frac{V}{I}$

$$= \frac{100}{10} = 10\Omega$$

$$\frac{\Delta R}{R} = \left(\frac{\Delta V}{V} + \frac{\Delta I}{I}\right)$$

$$\Delta R = \left(\frac{\Delta V}{V} + \frac{\Delta I}{I}\right) R$$

$$= \left(\frac{5}{100} + \frac{0.2}{10}\right) 10$$

$$= (0.05 + 0.02)10$$

$$= 0.07 \times 10 = 0.7$$

The resistance R = (10 ± 0.7)Ω

(v) Error in the power of a quantity

Consider the nth power of A, $Z = A^n$

The error ΔZ in Z is given by

$$Z \pm \Delta Z = (A \pm \Delta A)^n = A^n \left(1 \pm \frac{\Delta A}{A}\right)^n$$

$$= Z\left(1 \pm n\frac{\Delta A}{A}\right)$$

We get $[(1+x)^n \approx 1+nx$, when $x<<1]$ neglecting remaining terms, Dividing both sides by Z

$$1 \pm \frac{\Delta Z}{Z} = 1 \pm n\frac{\Delta A}{A} \; or$$

$$\frac{\Delta Z}{Z} = n.\frac{\Delta A}{A} \qquad (1.7)$$

The fractional error in the nth power of a quantity is n times the fractional error in that quantity.

General rule: If $Z = \frac{A^p B^q}{C^r}$ Then

maximum fractional error in Z is

given by $\frac{\Delta Z}{Z} = p\frac{\Delta A}{A} + q\frac{\Delta B}{B} + r\frac{\Delta C}{C}$

The percentage error in Z is given by

$$\frac{\Delta Z}{Z} \times 100 = p\frac{\Delta A}{A} \times 100 + q\frac{\Delta B}{B} \times 100 + r\frac{\Delta C}{C} \times 100$$

EXAMPLE 1.9

A physical quantity *x* is given by $x = \frac{a^2 b^3}{c\sqrt{d}}$. If the percentage errors of measurement in a, b, c and d are 4%, 2%, 3% and 1% respectively then calculate the percentage error in the calculation of *x*. (Related to NEET 2013)

Solution

Given $x = \frac{a^2 b^3}{c\sqrt{d}}$